

A Singularity-free Hierarchical Nonlinear Quad-Rotorcraft Control using Saturation and Barrier Lyapunov Function

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Abstract—A singularity-free nonlinear hierarchical control framework is proposed in this paper for control of a quad-rotorcraft unmanned aerial vehicle (UAV). A saturation control scheme with hyperbolic tangent function is designed for the position loop controller to ensure non-singular command attitude extraction and the effect of nonlinear coupling between the position and attitude subsystem is subsequently analyzed. The problem of sign-ambiguity commonly appears in reference attitude is overcome using arc tangent function by considering the signs of both the arguments. To obviate the problem of singularity during attitude tracking, a non-singular attitude loop controller using barrier Lyapunov function (BLF) with corresponding initial condition constraint is proposed. A rigorous stability analysis proves that the overall closed-loop system is asymptotically stable (AS) and all the signals are bounded in the cascaded control structure. The singularity-free hierarchical control development and its stability analysis exploits the full state space Euler-Lagrange under-actuated dynamics in terms of generalized coordinates with generalized force and torque collocated with generalized velocities. Simulation results show the performance of the proposed controller.

I. INTRODUCTION

It is well understood that the under-actuated quad-rotorcraft dynamics include a position and an attitude subsystem where a virtual control input is considered for the translational dynamics from which the command attitude and thrust is obtained. The hierarchical nonlinear control strategy [1]–[7], [19] requires desired attitude extraction for attitude control from the designed thrust input of the position subsystem. In [2], [3], [6] a multistage constructive procedure exploits the cascade property of the translational and rotational dynamics. A linear acceleration driving the outer loop toward the desired trajectory is designed from which a thrust and desired attitude is extracted to develop the required torque for driving the inner loop toward the command orientation. However, singularity is most likely to occur during extraction. Successive differentiation of virtual control input manifests a degeneracy problem if thrust boundedness is not considered and thereby fails to provide singularity-free reference attitude calculation. The widely used saturation functions [8]–[10] for position loop control design of the quad-rotorcraft are not smooth enough for the subsequent design of attitude control loop based on the

hierarchical strategy. Moreover, the attitude of the quad-rotorcraft is represented with Euler angles where singularity often occurs in the course of attitude tracking. Classical attitude controllers [11]–[13] are developed on the assumption that no such singularity happens. Quaternion based approach [14], [15] has been used to avoid the singularity problem of a quad-rotorcraft dynamics but applied only for attitude stabilization. At the same time, since the desired attitude is regarded as input to the rotational dynamics, the attitude tracking error introduces a nonlinear coupling between the two subsystems, whose effect on stability of the overall system is of utmost importance. In [16] the position and attitude loop control laws are designed based on smooth saturation and a barrier Lyapunov function (BLF) where a complete position and attitude tracking is achieved without singularity but the effect of attitude error on position tracking due to time-scale separation inherent to multi-rotor dynamics is not analyzed.

The proposed work is a singularity-free hierarchical nonlinear control design that uses Euler-Lagrange (E-L) dynamics of a quad-rotorcraft to compute guidance and control laws simultaneously. The E-L model considers generalized forces and torques collocated with generalized velocities and represents a full state-space under-actuated dynamics in terms of generalized coordinates. Earlier works on control hierarchy [1]–[7], [18] have pointed out the need, to use a smooth saturation function at the time of designing a virtual control input for position subsystem in order to avoid the singularity problem which manifests from successive differentiation during computation of reference attitude. It also suggests to impose a strict constraint on the design to ensure its vertical component should be strictly positive, otherwise desired pitch reaches $\pm \frac{\pi}{2}$ and generates the problem of attitude singularity while tracking the reference trajectory. In such a case, velocity transformation matrix degenerates and computed torque control input for attitude tracking becomes unbounded. Therefore an output constraint (pitch angle must be strictly less than $\pm \frac{\pi}{2}$) requires to be satisfied for all the time during attitude control to avoid singularity. In order to address both the challenges, the current outer loop position control is designed based on a hyperbolic tangent function of the filtered-tracking error whereas the inner attitude loop controller is developed using BLF [17] with a proposed constraint on output and initial condition. The saturated position and singularity-free attitude control laws simultaneously ensure non-singular extraction of desired attitude and trajectory tracking for the cascade system. The problem of sign-ambiguity commonly appears

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in reference attitude [1], [7], [16] calculation is overcome using arc tangent function by considering the signs of both the arguments. The Lyapunov stability analysis of the overall system proves that the origin of position and orientation error dynamics is asymptotically stable (AS) with the proposed singularity-free control algorithms applied on both the subsystems.

II. A QUAD-ROTORCRAFT MODEL- THE EULER-LAGRANGE APPROACH

The Lagrangian dynamics of a quad-rotorcraft is given by

$$M(q_v)\ddot{q}_v + V_m(q_v, \dot{q}_v)\dot{q}_v + G = H(\eta)F \quad (1)$$

where the pose $q_v = [\zeta^T \ \eta^T]^T \in \mathbb{R}^6$ of a quad-rotorcraft contains position and orientation vector (roll, pitch, and yaw angles) of the body-fixed frame $\{F_B\}$ relative to inertial frame $\{F_I\}$ expressed in $\{F_I\}$ i.e. $\zeta = [x \ y \ z]^T$ and $\eta = [\phi \ \theta \ \psi]^T$ as shown in Fig.1. $M(q_v) \in \mathbb{R}^{6 \times 6}$ represents the mass-inertia matrix, $V_m(q_v, \dot{q}_v) \in \mathbb{R}^{6 \times 6}$ represents the centripetal-coriolis matrix, $G \in \mathbb{R}^6$ represents the constant gravity vector are all expressed in $\{F_I\}$ whereas $F = [T \ \tau]^T \in \mathbb{R}^4$ represents the wrench (force $T \in \mathbb{R}$ and torque $\tau \in \mathbb{R}^3$) input vector expressed in $\{F_B\}$. Each motor M_i ($i=1(1)4$)¹ produces force F_i and $T = \sum_{i=1}^4 F_i \in \mathbb{R}$ represents total thrust applied on the vehicle. The roll-torque (τ_ϕ) is a differential function of ($F_2 \sim F_4$), the pitch-torque (τ_θ) is of ($F_1 \sim F_3$) and the yaw torque $\tau_\psi = \sum_{i=1}^4 \tau_{M_i} \in \mathbb{R}$ where τ_{M_i} is the reaction torque of motor M_i due to shaft acceleration and blades drag. Total input torque acting on the vehicle is represented by $\tau = [\tau_\phi \ \tau_\theta \ \tau_\psi]^T \in \mathbb{R}^3$. The matrix

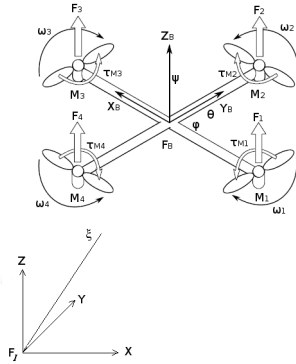


Fig. 1. Quad-Rotorcraft

$$H(\eta) \triangleq \begin{bmatrix} (c_\phi s_\theta c_\psi + s_\phi s_\psi) & 0 & 0 & 0 \\ (c_\phi s_\theta s_\psi - s_\phi c_\psi) & 0 & 0 & 0 \\ c_\phi c_\theta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & c_\phi & -s_\phi \\ 0 & -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix} \in \mathbb{R}^{6 \times 4}$$

is the mapping between the external forces/torques $F \in \mathbb{R}^4$ acting on the vehicle and the generalized wrench. Here $s_{(\cdot)}$ and $c_{(\cdot)}$ are defined as $s_{(\cdot)} \triangleq \sin(\cdot)$ and $c_{(\cdot)} \triangleq \cos(\cdot)$.

¹The notation "i=a(m)b" implies that "i=a,a+m,a+2m,...,b".

Equation (1) shows that quad-rotorcraft is an under-actuated mechanical system (UMS) having two connected subsystems 1) position subsystem and 2) attitude subsystem.

A. Position Subsystem

Using (1) the dynamics of the position subsystem is given by

$$M_q \ddot{\zeta} + \bar{G} = R(\eta)^T e_z T = T_v \quad (2)$$

where $M_q = m_q I \in \mathbb{R}^{3 \times 3}$ represents mass-matrix (m_q = mass of the quad-rotorcraft, $I \in \mathbb{R}^{3 \times 3}$, denotes an identity matrix), $\bar{G} = [0 \ 0 \ m_q g]^T \in \mathbb{R}^3$ represents the reduced order gravity vector, $R(\eta)$ is the rotation matrix between $\{F_B\}$ & $\{F_I\}$, e_z is the Cartesian basis vector of $\{F_I\}$ along z direction, and computed thrust control $T_v = [T_{v_x} \ T_{v_y} \ T_{v_z}]^T \in \mathbb{R}^3$ where T_{v_x} , T_{v_y} , T_{v_z} are the scalar components of T_v along XYZ axes. Equation (2) in matrix form is given by

$$\begin{bmatrix} m_q & 0 & 0 \\ 0 & m_q & 0 \\ 0 & 0 & m_q \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ m_q g \end{bmatrix} = \begin{bmatrix} (c_\phi s_\theta c_\psi + s_\phi s_\psi)T \\ (c_\phi s_\theta s_\psi - s_\phi c_\psi)T \\ c_\phi c_\theta T \end{bmatrix} \quad (3)$$

B. Attitude Subsystem

Using (1) the dynamics of the attitude subsystem is given by

$$M_\eta(\eta) \ddot{\eta} + V_\eta(\eta, \dot{\eta}) \dot{\eta} = J(\eta)^T \tau \quad (4)$$

where $M_\eta(\eta) \in \mathbb{R}^{3 \times 3}$ represents inertia tensor, $V_\eta(\eta, \dot{\eta}) \in \mathbb{R}^{3 \times 3}$ represents the reduced order centripetal-coriolis matrix,

$J(\eta) = \begin{bmatrix} 1 & 0 & -s_\theta \\ 0 & c_\phi & c_\theta s_\phi \\ 0 & -s_\phi & c_\theta c_\phi \end{bmatrix}$ represents velocity transformation matrix.

Property 1. The matrix $[\dot{M}_\eta(\eta) - 2V_\eta(\eta, \dot{\eta})] \in so(3)$ ² i.e. $[\dot{M}_\eta(\eta) - 2V_\eta(\eta, \dot{\eta})]^T = -[\dot{M}_\eta(\eta) - 2V_\eta(\eta, \dot{\eta})]$ and therefore, $v^T [\dot{M}_\eta(\eta) - 2V_\eta(\eta, \dot{\eta})] v = 0 \ \forall v \in \mathbb{R}^3$

From (2) and (4) the respective terms of the matrices in (1) are represented as follows

$$M(q_v) = \begin{bmatrix} M_q & 0^{3 \times 3} \\ 0^{3 \times 3} & M_\eta(\eta) \end{bmatrix}, \quad V_m(q_v, \dot{q}_v) = \begin{bmatrix} 0^{3 \times 3} & 0^{3 \times 3} \\ 0^{3 \times 3} & V_\eta(\eta, \dot{\eta}) \end{bmatrix},$$

$$G = [\bar{G} \ 0 \ 0 \ 0]^T \text{ and } H(\eta) = \begin{bmatrix} R(\eta)^T e_z & 0^{3 \times 3} \\ 0^{3 \times 1} & J(\eta)^T \end{bmatrix}.$$

The following section proposes a nonlinear hierarchical control law design for attitude and position subsystem.

III. DESIGN OF CONTROL HIERARCHY

The objective is to design a flight controller by exploiting the structural properties of rotorcraft model to simultaneously

²Often denoted as a group of 3×3 skew-symmetric matrix

control the position and orientation dynamics of the quadrotor. A hierarchical nonlinear control is used where the inner-loop performs attitude tracking and generates the required torques whereas the outer-loop generates the thrust and the reference attitude, required to follow a desired trajectory $\zeta_d = [x_d \ y_d \ z_d]^T$. The developed thrust control input is bounded to ensure singular-free extraction of reference roll and pitch of the quad-rotorcraft i.e. $\{\phi_d(t), \theta_d(t)\} \in (-\frac{\pi}{2}, \frac{\pi}{2})$, $\forall t \geq 0$ whereas a BLF based attitude control law is designed to ensure pitch angle $\theta(t) \in (-\frac{\pi}{2}, \frac{\pi}{2})$. The problem of sign-ambiguity appears in reference attitude is overcome using arc tangent function by considering the signs of both the arguments.

A. Position Control Design

The objective is to develop a position tracking controller for the quad-rotorcraft dynamics given by (2). The position error is defined as $e_\zeta(t) = \zeta_d - \zeta = [e_x \ e_y \ e_z]^T \in \mathbb{R}^3$. The order of the dynamic expression given in (2) is subsequently reduced to facilitate the control development by defining a filtered tracking error-like variable $r_\zeta(t) \in \mathbb{R}^3$ as

$$r_\zeta = \dot{e}_\zeta + \alpha_\zeta \tanh(e_\zeta) + \tanh(e_f) \quad (5)$$

where $\alpha_\zeta \in \mathbb{R}_+$ represents a filter gain and to aid the subsequent control design and analysis, the vector, matrix functions $\tanh(\cdot)$, $\cosh(\cdot)$, $\text{sech}(\cdot)$ are defined as

$$\tanh(v) \triangleq [\tanh(v_1) \ \tanh(v_2) \ \tanh(v_3)]^T \in \mathbb{R}^3,$$

$$\cosh(v) \triangleq \text{diag}\{\cosh(v_1), \cosh(v_2), \cosh(v_3)\} \text{ and}$$

$$\text{sech}(v) \triangleq \cosh^{-1}(v) \in \mathbb{R}^{3 \times 3} \text{ where } v = [v_1 \ v_2 \ v_3]^T \in \mathbb{R}^3.$$

An auxiliary filter variable $e_f \triangleq [e_{f_x} \ e_{f_y} \ e_{f_z}]^T \in \mathbb{R}^3$ is defined to have the following dynamics

$$\dot{e}_f = \cosh^2(e_f)(-k_\zeta r_\zeta + \tanh(e_\zeta) - \gamma_\zeta \tanh(e_f)) \quad (6)$$

where $e_f(0) = 0$ and $\{k_\zeta, \gamma_\zeta\} \in \mathbb{R}_+$ are constants.

Remark 1. The choice of filter variables in (5) and (6) are instrumental to provide singularity-free reference attitude calculation.

Taking the time derivative of (5), pre-multiplying the resulting expression by M_q , substituting (2) and second time derivative of $e_\zeta(t)$, the open-loop position error system for $r_\zeta(t)$ is formulated as

$$\begin{aligned} M_q \dot{r}_\zeta &= M_q \ddot{\zeta}_d - M_q k_\zeta r_\zeta + M_q \tanh(e_\zeta) + \alpha_\zeta M_q \text{sech}^2(e_\zeta) \dot{e}_\zeta \\ &\quad - \gamma_\zeta M_q \tanh(e_f) + \bar{G} - T_v \\ &= M_q \ddot{\zeta}_d - M_q k_\zeta r_\zeta + M_q \tanh(e_\zeta) + \chi + \bar{G} - T_v \end{aligned} \quad (7)$$

where $\chi \triangleq \alpha_\zeta M_q \text{sech}^2(e_\zeta) \dot{e}_\zeta - \gamma_\zeta M_q \tanh(e_f)$. Assume the open-loop position error system can be stabilized by a smooth virtual control law T_d . Adding and subtracting T_d on the right-hand side of (7), the equivalent representation of open-loop position error dynamics is obtained as

$$\begin{aligned} M_q \dot{r}_\zeta &= M_q \ddot{\zeta}_d - M_q k_\zeta r_\zeta + M_q \tanh(e_\zeta) + \chi + \bar{G} \\ &\quad - (T_v - T_d) - T_d \\ &= M_q \ddot{\zeta}_d - M_q k_\zeta r_\zeta + M_q \tanh(e_\zeta) + \chi + \bar{G} - T_e - T_d \end{aligned} \quad (8)$$

where $T_e \triangleq T_v - T_d$ denotes the error between actual and desired thrust.

Remark 2. Quad-rotorcraft dynamics has additional position control input which differs from classical back-stepping design where there is one control and rest are virtual inputs.

Based on the resulting system in (8), T_d is designed as

$$T_d = M_q \ddot{\zeta}_d + 2M_q \tanh(e_\zeta) - k_\zeta M_q \tanh(e_f) + \bar{G} \quad (9)$$

assuming the dynamics is exactly known and $k_\zeta \in \mathbb{R}_+$ represents a control gain. Substituting (9) into (8), the closed-loop error dynamics is given by

$$M_q \dot{r}_\zeta = -M_q k_\zeta r_\zeta - M_q \tanh(e_\zeta) + k_\zeta M_q \tanh(e_f) + \chi - T_e \quad (10)$$

B. Attitude Reference Calculation

Using (3) the following set of equations are derived as

$$m_q \ddot{x} = (c_\phi s_\theta c_\psi + s_\phi s_\psi) T \quad (11)$$

$$m_q \ddot{y} = (c_\phi s_\theta s_\psi - s_\phi c_\psi) T \quad (12)$$

$$m_q \ddot{z} + m_q g = c_\phi c_\theta T \quad (13)$$

From above equations the scalar components of $T_d = [T_{x_d} \ T_{y_d} \ T_{z_d}]^T \in \mathbb{R}^3$ on $\{X, Y, Z\}$ axes, are defined as

$$T_{x_d} \triangleq (c_\phi s_\theta c_\psi_d + s_\phi s_\psi_d) T \quad (14)$$

$$T_{y_d} \triangleq (c_\phi s_\theta s_\psi_d - s_\phi c_\psi_d) T \quad (15)$$

$$T_{z_d} \triangleq c_\phi c_\theta_d T \quad (16)$$

where ϕ_d, θ_d, ψ_d are the desired attitude of the vehicle and

$$\|T_d\| = \sqrt{T_{x_d}^2 + T_{y_d}^2 + T_{z_d}^2} = T \quad (17)$$

Using (14), (15), and (16) the desired roll and pitch angles (ϕ_d, θ_d) are computed as

$$\phi_d = \arctan((T_{x_d} s_{\psi_d} - T_{y_d} c_{\psi_d}) / \sqrt{(T_{x_d} c_{\psi_d} + T_{y_d} s_{\psi_d})^2 + T_{z_d}^2}), \quad \phi_d \in (-\frac{\pi}{2}, \frac{\pi}{2}) \quad (18)$$

$$\theta_d = \arctan((T_{x_d} c_{\psi_d} + T_{y_d} s_{\psi_d}) / T_{z_d}), \quad \theta_d \in (-\frac{\pi}{2}, \frac{\pi}{2}) \quad (19)$$

They are given as reference inputs to the attitude subsystem for any user-supplied fixed yaw angle ψ_d .

Proposition 1. A stricter requirement should be imposed on the gain parameter of position loop controller in order to ensure $T_{z_d} \neq 0$, $\forall t \geq 0$ for non-singular extraction of reference roll and pitch of the quad-rotorcraft. The desired acceleration $\ddot{\zeta}_d = [\ddot{x}_d \ \ddot{y}_d \ \ddot{z}_d]^T$ must be bounded and satisfies the condition $|\ddot{z}_d| \leq \varepsilon < g-2$, $t \geq 0$, where ε is a positive scalar quantity. It implies that the magnitude of the Z-component of desired acceleration should be less than $g-2$. Moreover to ensure the desired thrust is strictly positive, the controller parameter k_ζ satisfies the following constraint

$$k_\zeta < g - \varepsilon - 2 \quad (20)$$

Extracting the third row of (9) and using (20)

$$T_{z_d} = m_q \ddot{z}_d + 2m_q \tanh(e_z) - k_\zeta m_q \tanh(e_{f_z}) + m_q g \quad (21)$$

the mass normalized vertical thrust control input,

$$\begin{aligned} u_{z_d} &= \ddot{z}_d + 2 \tanh(e_z) - k_\zeta \tanh(e_{f_z}) + g \\ &\geq -\varepsilon - 2 - k_\zeta + g > 0 \end{aligned} \quad (22)$$

u_{z_d} is strictly positive $\forall t \geq 0$. In this case, the command roll ϕ_d and pitch θ_d can be extracted via (18) and (19) without singularity.

C. Singularity-free Attitude Control Design

The objective is to develop an output constrained attitude tracking controller for the quad-rotorcraft dynamics given by (4). The attitude error is defined as $e_\eta(t) = \eta_d - \eta = [e_\phi \ e_\theta \ e_\psi]^T \in \mathbb{R}^3$ where $\eta_d(t) = [\phi_d(t) \ \theta_d(t) \ \psi_d(t)]^T \in \mathbb{R}^3$ denotes the desired attitude. The order of the dynamic expression given in (4) is subsequently reduced to facilitate control development by defining a filtered tracking error-like variable $r_\eta(t) \in \mathbb{R}^3$

$$r_\eta = \dot{e}_\eta + A_\eta e_\eta \quad (23)$$

where $A_\eta = \text{diag}\{\alpha_{\eta 11}, \alpha_{\eta 22}, \alpha_{\eta 33}\} \in \mathbb{R}^{3 \times 3}$, $\alpha_{\eta ii} \in \mathbb{R}_+$, $i=1(1)3$ is a positive-definite constant control gain matrix. Taking the time derivative of (23), pre-multiplying the resulting expression by $M_\eta(\eta)$, substituting (4) and second time derivative of $e_\eta(t)$, the open-loop attitude error system for $r_\eta(t)$ is formulated as

$$M_\eta(\eta) \dot{r}_\eta = M_\eta(\eta) \ddot{\eta}_d + M_\eta(\eta) A_\eta \dot{e}_\eta + V_\eta(\eta, \dot{\eta}) \dot{\eta} - J(\eta)^T \tau \quad (24)$$

Based on the open-loop attitude error system in (24) a computed torque control τ is designed as

$$\begin{aligned} \tau &= J(\eta)^{-T} (M_\eta(\eta) \ddot{\eta}_d + M_\eta(\eta) A_\eta \dot{e}_\eta + V_\eta(\eta, \dot{\eta}) \dot{\eta} + \\ &V_\eta(\eta, \dot{\eta}) r_\eta + K_\eta r_\eta + e_\eta + \bar{\eta}) \end{aligned} \quad (25)$$

assuming the dynamics is exactly known, $K_\eta \in \mathbb{R}^{3 \times 3}$ represents a positive diagonal control gain matrix and $\bar{\eta} \triangleq [e_\phi \ e_\theta/(k_\theta^2 - e_\theta^2) \ e_\psi]^T \in \mathbb{R}^3$ where the attitude control gain $k_\theta(t) \in \mathbb{R}$ is chosen such that

$$k_\theta(t) = \pi/2 - |\theta_d(t)| \quad (26)$$

and $|e_\theta(0)| < k_\theta(0)$. Substituting (25) into (24) the closed-loop error system is determined as

$$M_\eta(\eta) \dot{r}_\eta + V_\eta(\eta, \dot{\eta}) r_\eta + K_\eta r_\eta + e_\eta + \bar{\eta} = 0 \quad (27)$$

D. Attitude Error Characterization

Equation (8) indicates that T_v can be realized with certain errors. Using (3), (11), (12), (13), and $e_\eta(t)$ expressions for T_{v_x} , T_{v_y} , and T_{v_z} are as follows

$$T_{v_x} = (c_{\phi_d - e_\phi} s_{(\theta_d - e_\theta)} c_{(\psi_d - e_\psi)} + s_{(\phi_d - e_\phi)} s_{(\psi_d - e_\psi)}) T \quad (28)$$

$$T_{v_y} = (c_{(\phi_d - e_\phi)} s_{(\theta_d - e_\theta)} s_{(\psi_d - e_\psi)} - s_{(\phi_d - e_\phi)} c_{(\psi_d - e_\psi)}) T \quad (29)$$

$$T_{v_z} = c_{(\phi_d - e_\phi)} c_{(\theta_d - e_\theta)} T \quad (30)$$

Using trigonometric identities³ (28) can be rewritten as

$$\begin{aligned} T_{v_x} &= (c_{\phi_d} s_{\theta_d} c_{\psi_d} + s_{\phi_d} s_{\psi_d}) T + (2(s_{\theta_d} c_{\psi_d} s_{(\phi_d - e_{\phi/2})} s_{e_{\phi/2}} \\ &- c_{\phi_d} c_{\psi_d} s_{(\theta_d - e_{\theta/2})} s_{e_{\theta/2}} + c_{\phi_d} s_{\theta_d} s_{(\psi_d - e_{\psi/2})} s_{e_{\psi/2}}) \\ &- 4(c_{\psi_d} s_{(\phi_d - e_{\phi/2})} c_{(\theta_d - e_{\theta/2})} s_{e_{\phi/2}} s_{e_{\theta/2}} - s_{\theta_d} s_{(\phi_d - e_{\phi/2})} s_{(\psi_d - e_{\psi/2})} s_{e_{\phi/2}} s_{e_{\psi/2}} \\ &+ c_{\phi_d} c_{(\theta_d - e_{\theta/2})} s_{(\psi_d - e_{\psi/2})} s_{e_{\theta/2}} s_{e_{\psi/2}}) \\ &- 8s_{(\phi_d - e_{\phi/2})} c_{(\theta_d - e_{\theta/2})} s_{(\psi_d - e_{\psi/2})} s_{e_{\phi/2}} s_{e_{\theta/2}} s_{e_{\psi/2}}) T \\ &= (c_{\phi_d} s_{\theta_d} c_{\psi_d} + s_{\phi_d} s_{\psi_d}) T + h_x(\phi_d, \theta_d, \psi_d, e_\phi, e_\theta, e_\psi) T \\ &= T_{x_d} + h_x(\eta_d, e_\eta) T \end{aligned} \quad (31)$$

Similarly (29) and (30) reduces to

$$T_{v_y} = T_{y_d} + h_y(\eta_d, e_\eta) T \quad (32)$$

$$T_{v_z} = T_{z_d} + h_z(\eta_d, e_\eta) T \quad (33)$$

where h_x , h_y and h_z are the nonlinear coupling terms between the position and attitude subsystem, consisting of sine and cosine functions of the desired attitude and their errors. The computed thrust control T_v is designed as

$$T_v - T_d = T_e = T h = \|T_d\| h \quad (34)$$

where $h = [h_x \ h_y \ h_z]^T \in \mathbb{R}^3$

IV. STABILITY ANALYSIS

Stability of the complete system is analyzed using a hierarchical control design in presence of a strong nonlinear coupling between the two subsystems.

Lemma 1. Based on the definition of (26) and considering bounded thrust derivatives $|\dot{k}_\theta/k_\theta| \leq v$ where v is a finite bound⁴ on the interval $\theta_d(t) \in (-\frac{\pi}{2}, \frac{\pi}{2})$

Lemma 2. For a bounded second derivative of position reference $\zeta_d(t)$, using (9) $\|T_d\| \leq c$ where c is a positive scalar and defined as, $c \triangleq (c_1 + c_2)$, $c_1 \triangleq (k_1 k_2 + k_3)$, $c_2 \triangleq (k_\zeta + 2) k_1 \sqrt{3}$, $k_1 \triangleq \|M_q\|_F$, $k_2 \triangleq \|\ddot{\zeta}_d\|$, $k_3 \triangleq \|\ddot{G}\|$

It is noted that $\|\cdot\|_F$ denotes Frobenius norm of the argument matrix and it is derived from (9) using (5) and subsequently applying norm inequalities.

Lemma 3. $\|h\| \leq \alpha \|e_\eta\|$, where $e_\eta(t)$ is the attitude-tracking error vector and α is a positive constant.

It is derived from the expressions of $h_x(\eta_d, e_\eta)$, $h_y(\eta_d, e_\eta)$ and $h_z(\eta_d, e_\eta)$ appear in (31), (32) and (33) and applying trigonometric identities⁵ and norm inequalities.

³The following trigonometric identities are applied to obtain (31), (32) and (33) respectively where $\{x, y\} \in \mathbb{R}$.

$$\begin{aligned} \cos(x - y) &= \cos(x) + 2 \sin(x - y/2) \sin(y/2) \\ \sin(x - y) &= \sin(x) - 2 \cos(x - y/2) \sin(y/2) \end{aligned}$$

⁴It can be ensured through the strictly less than condition in (20)

⁵The following inequalities are applied to prove the lemma.

$$\begin{aligned} |\sin(x)|, |\cos(x)| &\leq 1 \\ |\sin(x)| &\leq |x| \quad \forall x \in (-\pi/2, \pi/2) \\ |\sin(x - \frac{y}{2}) \sin(\frac{y}{2})|, |\cos(x - \frac{y}{2}) \sin(\frac{y}{2})| &\leq |\sin(\frac{y}{2})| \end{aligned}$$

Lemma 4. The term χ can be upper bounded as $\|\chi\| \leq \delta \|z_\zeta\|$, where $z_\zeta \triangleq \begin{bmatrix} r_\zeta^T & \text{Tanh}^T(e_\zeta) & \text{Tanh}^T(e_f) \end{bmatrix}^T \in \mathbb{R}^9$ and δ is a positive constant.

It is derived from the previous definition of χ term and subsequently using (6) and norm inequalities. The above lemmas⁶ will be instrumental in proving the main theorem.

Theorem 5. For the system in (1), the position control law in (9) with the attitude control law in (25) guarantee that the overall error dynamics $e_v(t) = \begin{bmatrix} e_\zeta^T & e_\eta^T \end{bmatrix}^T$ is asymptotically stable provided the following gain conditions are satisfied

$$\begin{aligned} k\kappa_{\zeta_2} &> \delta^2/4\lambda_{\min}\{M_q\} \\ \lambda_{\min}\{A_\eta\} &> \max\{\beta^2/4\kappa_{\zeta_3}\lambda_{\min}\{M_q\}, 2v\} \end{aligned} \quad (35)$$

where $\lambda_{\min}$ represents minimum eigenvalue of the argument matrix and $(k, \kappa_{\zeta_2}, \kappa_{\zeta_3}, \beta)$ are subsequently defined auxiliary positive scalar constants subjected to an initial condition constraint $|e_\theta(0)| < k_\theta(0)$.

Proof. Consider a Lyapunov function candidate as

$$\begin{aligned} L(t) &= L_\zeta(t) + L_\eta(t) \\ &= (1/2r_\zeta^T M_q r_\zeta + m_q (\ln \cosh(e_x) + \ln \cosh(e_y) + \ln \cosh(e_z)) \\ &\quad + 1/2 \text{Tanh}^T(e_f) M_q \text{Tanh}(e_f)) + (1/2r_\eta^T M_\eta(\eta) r_\eta + 1/2e_\eta^T e_\eta \\ &\quad + 1/2(e_\phi^2 + \ln(k_\theta^2/(k_\theta^2 - e_\theta^2)) + e_\psi^2)) \end{aligned} \quad (36)$$

For simplicity consider $L_\zeta(t)$ at first

$$\begin{aligned} L_\zeta(t) &= 1/2r_\zeta^T M_q r_\zeta + m_q (\ln \cosh(e_x) + \ln \cosh(e_y) \\ &\quad + \ln \cosh(e_z)) + 1/2 \text{Tanh}^T(e_f) M_q \text{Tanh}(e_f) \end{aligned} \quad (37)$$

Differentiate (37), substitute (6) and (10) and canceling like terms, results

$$\begin{aligned} \dot{L}_\zeta(t) &= -k_\zeta r_\zeta^T M_q r_\zeta - r_\zeta^T T_e - \alpha_\zeta \text{Tanh}^T(e_\zeta) M_q \text{Tanh}(e_\zeta) \\ &\quad - \gamma_\zeta \text{Tanh}^T(e_f) M_q \text{Tanh}(e_f) + r_\zeta^T \chi \end{aligned} \quad (38)$$

Again consider $L_\eta(t)$

$$L_\eta(t) = 1/2r_\eta^T M_\eta(\eta) r_\eta + 1/2e_\eta^T e_\eta + 1/2(e_\phi^2 + \ln(k_\theta^2/(k_\theta^2 - e_\theta^2)) + e_\psi^2) \quad (39)$$

Differentiate (39), substitute (23) and (27), apply Property 1, and canceling like terms, results

$$\dot{L}_\eta(t) = -r_\eta^T K_\eta r_\eta - e_\eta^T A_\eta e_\eta - \bar{\eta}^T A_\eta e_\eta - k_\theta e_\theta^2/k_\theta(k_\theta^2 - e_\theta^2) \quad (40)$$

Add (38) and (40), use (34), apply Lemma 1-4, and subsequently upper bound the resulting expression yields

$$\begin{aligned} \dot{L}(t) &\leq -k_\zeta \lambda_{\min}\{M_q\} \|r_\zeta\|^2 \\ &\quad - \alpha_\zeta \lambda_{\min}\{M_q\} \|\text{Tanh}(e_\zeta)\|^2 - \gamma_\zeta \lambda_{\min}\{M_q\} \|\text{Tanh}(e_f)\|^2 \\ &\quad - \lambda_{\min}\{K_\eta\} \|r_\eta\|^2 - \lambda_{\min}\{A_\eta\} \|e_\eta\|^2 \\ &\quad - \lambda_{\min}\{A_\eta\} (e_\phi^2 + e_\theta^2/(k_\theta^2 - e_\theta^2) + e_\psi^2) \\ &\quad + v/(k_\theta^2 - e_\theta^2) e_\theta^2 + \|r_\zeta\| \alpha_\zeta \|e_\eta\| + \|r_\zeta\| \delta \|z_\zeta\| \end{aligned} \quad (41)$$

⁶The detailed proofs of Lemma 1–4 are not provided here to honor the page limit and will be reported in the future publication of the authors.

Define $k_\zeta \triangleq (\kappa_{\zeta_1} + \kappa_{\zeta_2} + \kappa_{\zeta_3})$, $\beta \triangleq \alpha_\zeta$, and completing the squares, (41) becomes

$$\begin{aligned} \dot{L}(t) &\leq -k(\|r_\zeta\|^2 + \|\text{Tanh}(e_\zeta)\|^2 + \|\text{Tanh}(e_f)\|^2) \\ &\quad - k_{\zeta_2} \lambda_{\min}\{M_q\} (\|r_\zeta\| - \delta \|z_\zeta\|/2k_{\zeta_2} \lambda_{\min}\{M_q\})^2 \\ &\quad + \delta^2 \|z_\zeta\|^2 / 4k_{\zeta_2} \lambda_{\min}\{M_q\} \\ &\quad - k_{\zeta_3} \lambda_{\min}\{M_q\} (\|r_\zeta\| - \beta \|e_\eta\|/2k_{\zeta_3} \lambda_{\min}\{M_q\})^2 \\ &\quad + \beta^2 \|e_\eta\|^2 / 4k_{\zeta_3} \lambda_{\min}\{M_q\} \\ &\quad - \lambda_{\min}\{K_\eta\} \|r_\eta\|^2 - \lambda_{\min}\{A_\eta\} \|e_\eta\|^2 \\ &\quad - \lambda_{\min}\{A_\eta\} (e_\phi^2 + e_\theta^2/(k_\theta^2 - e_\theta^2) + e_\psi^2) + v/(k_\theta^2 - e_\theta^2) e_\theta^2 \end{aligned} \quad (42)$$

where $k \triangleq \min(k_{\zeta_1} \lambda_{\min}\{M_q\}, \alpha_\zeta \lambda_{\min}\{M_q\}, \gamma_\zeta \lambda_{\min}\{M_q\})$. Further upper bounding on (42) results

$$\begin{aligned} \dot{L}(t) &\leq -(k - \delta^2/4k_{\zeta_2} \lambda_{\min}\{M_q\}) \|z_\zeta\|^2 - \lambda_{\min}\{K_\eta\} \|r_\eta\|^2 \\ &\quad - (\lambda_{\min}\{A_\eta\} - \beta^2/4k_{\zeta_3} \lambda_{\min}\{M_q\}) \|e_\eta\|^2 \\ &\quad - \lambda_{\min}\{A_\eta\} / 2(k_\theta^2 - e_\theta^2) (1 - 2v/\lambda_{\min}\{A_\eta\}) e_\theta^2 \\ &\quad - \lambda_{\min}\{A_\eta\} (e_\phi^2 + e_\theta^2/2(k_\theta^2 - e_\theta^2) + e_\psi^2) \end{aligned} \quad (43)$$

Imposing the gain condition (35) on (43) yields

$$\begin{aligned} \dot{L}(t) &\leq -k_a \|z_\zeta\|^2 - \lambda_{\min}\{K_\eta\} \|r_\eta\|^2 \\ &\quad - \min(k_b, \lambda_{\min}\{A_\eta\} k_c / 2(k_\theta^2 - e_\theta^2)) e_\theta^2 \\ &\quad - \lambda_{\min}\{A_\eta\} (e_\phi^2 + e_\theta^2/(k_\theta^2 - e_\theta^2) + e_\psi^2) \end{aligned} \quad (44)$$

negative-definite (ND) result and the error dynamics is AS where $k_a \triangleq (k - \delta^2/4k_{\zeta_2} \lambda_{\min}\{M_q\}) > 0$, $k_b \triangleq (\lambda_{\min}\{A_\eta\} - \beta^2/4k_{\zeta_3} \lambda_{\min}\{M_q\}) > 0$ and $k_c \triangleq (1 - 2v/\lambda_{\min}\{A_\eta\}) > 0$ In view of (36)

$$\begin{aligned} 1/2 \ln(k_\theta^2/(k_\theta^2 - e_\theta^2)) &\leq L(0) < \infty \\ \implies (k_\theta^2/(k_\theta^2 - e_\theta^2)) &\leq e^{2L(0)} \implies e_\theta^2/k_\theta^2 \leq 1 - e^{-2L(0)} \\ \text{i.e. } e_\theta^2/k_\theta^2 < 1 &\implies |e_\theta(t)| < k_\theta(t) \quad \forall t \geq 0 \end{aligned} \quad (45)$$

Using $e_\eta(t)$ and applying the definition (26), $\forall t \geq 0$

$$|\theta(t)| \leq |\theta_d(t)| + |e_\theta(t)| < |\theta_d(t)| + k_\theta(t) = \pi/2 \quad (46)$$

Remark 3. With the help of BLF, (45) proves that $e_\theta(t)$ is strictly less than $k_\theta(t)$ for all the time provided the initial condition constraint is satisfied. Using above result, (46) restricts pitch angle $|\theta(t)| < \pm\pi/2$ and singularity is avoided.

V. SIMULATION RESULTS

The control hierarchy developed in presence of nonlinear coupling between position and attitude subsystem of a quadrotorcraft has been tested on a simulated environment. The list of model parameters used to verify the design are $m_q = 468\text{g}$, moment of inertia along $\{X_B, Y_B, Z_B\}$ axes are $I_{xx} = 4.9$, $I_{yy} = 4.9$ and $I_{zz} = 8.8\text{g.m}^2$. Control and filter gains are $k_\zeta = \alpha_\zeta = \gamma_\zeta = 1$, and $K_\eta = A_\eta = \text{diag}(100, 100, 100)$. The reference trajectory is defined as $x_d = \cos(t)$, $y_d =$

$\sin(t)$, $z_d = t^7$ and $\psi_d = 0$. Fig. 2 and 3 exhibits position and attitude tracking performance of the controller. Result shows singularity-free reference pitch extraction and actual pitch lies in $\{(-\pi/2, \pi/2) - \text{blue line}\}$, during attitude tracking. Thrust components on $\{X, Y, Z\}$ axes are shown in Fig. 4.

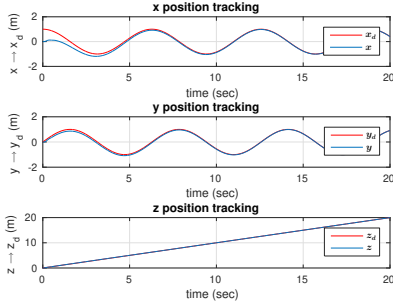


Fig. 2. Position Tracking

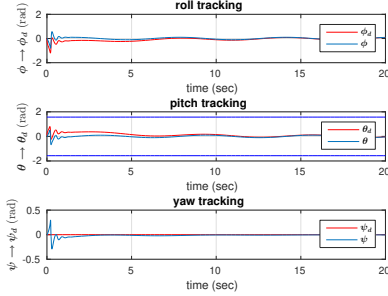


Fig. 3. Attitude Tracking

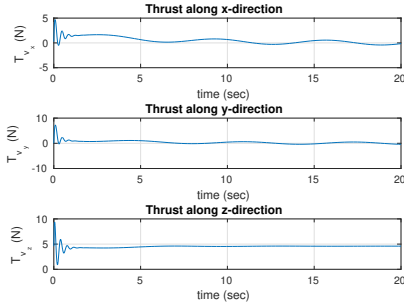


Fig. 4. Thrust along $\{X, Y, Z\}$ axes

VI. CONCLUSIONS

A singularity-free hierarchical control is proposed for trajectory tracking of a quad-rotorcraft. A saturation control using hyperbolic tangent function is designed for the position loop controller to ensure non-singular command attitude extraction. An attitude loop controller using BLF satisfying output and initial condition constraints is proposed to eliminate singularity problem during attitude tracking. The effect of nonlinear coupling between the position and attitude

⁷Mathematically it is unbounded but in practice the vehicle will always be at altitude lock position after arising a certain height.

subsystem is considered and the rigorous analysis proves that the overall closed-loop system is asymptotically stable. The design can be used in any quadrotor [20] applications.

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